



## Full Length Article

# Experimental study on combustion and emission performance of a spark-ignition engine fueled with water containing acetone-gasoline blends



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## HIGHLIGHTS

- Water containing acetone (WA)-gasoline blends with various water content were tested.
- Unregulated emissions of WA-gasoline blends were investigated using GC/MS and GC/FID.
- WA-gasoline blends could advance combustion phasing and enhance thermal efficiency.
- WA-gasoline blends could reduce the CO, NO<sub>x</sub> and aromatic hydrocarbon emissions.

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## ABSTRACT

Acetone is widely recognized as a major organic solvent and also a basic ingredient for varnishes and paints. In comparison with ethanol, acetone has more attractive properties as a potential alternative fuel such as its higher energy density, heating value, and volatility. However, the water content in acetone production would lead to a high cost for dehydration, and may prohibit acetone from being widely used in internal combustion engines. This has prompted research on using water containing acetone as a potential alternative fuel. In this study, various acetone-gasoline blends such as A10 (10 vol% acetone and 90 vol% gasoline by volume), A20 (20 vol% acetone and 80 vol% gasoline by volume), A19.5W0.5 (19.5 vol% acetone and 0.5 vol% water and 80 vol% gasoline) and A19W1 (19 vol% acetone and 1 vol% water and 80 vol% gasoline) were used as fuels in a port-fuel injection (PFI) spark ignition (SI) engine. The performance of test fuels was compared with that of G100 (gasoline) under various equivalence ratios ( $\Phi$ ) from 0.83 to 1.25 and at engine loads of 3 and 5 bar BMEP. In addition, aromatic compound emissions such as benzene, toluene, ethylbenzene, xylene (BTEX) were also measured by gas chromatograph (GC) with mass spectrometer and flame ionization detector. The results showed that A19W1 generally had a more advanced combustion phasing, higher brake thermal efficiency, and lower carbon monoxide (CO), nitrogen oxide (NO<sub>x</sub>) and BTEX than those of other test fuels. Therefore, water containing acetone-gasoline blends could be used as a good alternative fuel due to the improvement of engine performance and reduction of emissions.

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## 1. Introduction

During the past few decades, an increased consumption of fossil fuels by the transportation industry has exacerbated the oil crisis

and environment problems. Among various environment pollutants, automobile exhaust is one of major sources of atmospheric pollution. Typically, regulated emissions such as particulate matter (PM), nitrogen oxide (NO<sub>x</sub>), sulphur dioxide (SO<sub>2</sub>), carbon monoxide (CO) and unregulated emissions such as polycyclic aromatic hydrocarbons (PAHs), aldehydes, acids are the main components in vehicle exhaust gas [1–6]. Therefore, more considerations have been taken into the utilization of alternative fuels in engines.

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